Bayesian Mixture Model Analysis of Running Activity Distances

An assignment for MATH574 Bayesian Computational Statistics at Illinois Tech
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Table of Contents

Table of Contents 1
1. Introduction 2
2. Methodology 2
   2.1. Model Selection 2
   2.2. Implementation and Priors 2
   2.3. Data Preprocessing 3
3. Results 4
   3.1. Convergence and Parameter Estimation 4
   3.2. Prior Sensitivity Analysis 5
   3.3. Posterior Predictive Checks 7
   3.4. Classification and Prediction 7
   3.5. Training Timeline 8
4. Discussion and Conclusions 9
   4.1. Implications of Findings 9
   4.2. Suitability of the Bayesian Mixture Model 9
   4.3. Concerns and Reliability 9
5. Bibliography 10

1. Introduction

As a regular runner who sometimes prepares for marathons, the author noticed that his running tended to operate in two regimes - one where routine runs were completed mainly limited by being a convenient length of time (<1hr), and another where the length of runs varied far more as part of marathon training programs.

This report details the application of a Bayesian finite mixture model to analyze the authors running activity data from the fitness tracking platform Strava. A mixture model was chosen to fit with the two regime model described above - a “routine” run that was very common about 9km, and a more varied approach to training distances that occurred when following a training plan for a marathon or similar.

The dataset consists of N=220 filtered 'Run' activities, and it is aimed to model them using a two gaussian mixture model, consisting of a narrow peak representing the “routine” runs, and a much broader curve representing the “training” runs.

While one objective is to describe the mean and variance of the two activity regimes, it also could be useful to find a threshold that indicates a deviation from the routine regime to the training regime.

Note the Jupyter Notebook “*Project 3 - Strava.ipynb*” contains all R code relating to this report, and the file “*mixture_model.stan*” contains the stan model used for fitting.

2. Methodology

2.1. Model Selection

A two-component Normal finite mixture model was selected as the appropriate Bayesian technique to model the data, Y i.e. the distances of each run. This model posits that each observation y_i is generated by one of two components, $k \in \{1, 2\}$, with a corresponding mixing proportion π:

$$p(y_i \mid \theta) = \pi f_1(y_i \mid \mu_1, \sigma_1) + (1 - \pi) f_2(y_i \mid \mu_2, \sigma_2)$$

Where f_k are Normal distributions parameterized by mean μ_k and standard deviation σ_k, and routine runs correspond to $k = 1$ and training runs to $k = 2$.

2.2. Implementation and Priors

The model was implemented and fitted using the rstan package, utilizing Markov Chain Monte Carlo (MCMC) sampling, in a Jupyter notebook using R.

- **Priors:** Informative priors were set based on typical running habits: $\mu_1 \sim Normal(10, 1.5)$ (in km) for routine runs and $\mu_2 \sim Normal(25, 2)$ for training runs. Weakly informative Half-Cauchy priors were used for the standard deviations, σ_k, with scale parameters of 2 for routine runs and 4 for the broader distribution of the training runs.

A Beta(1,1) distribution was used as a weakly informative prior for π.

- **Identifiability Constraints:** To prevent label switching (where the component labels μ_1/σ_1 and μ_2/σ_2 switch places between MCMC iterations), two constraints were applied:
    1. **Variance Ordering:** $\sigma_1 < \sigma_2$, ensuring Component 1 is always the lower-variance, tighter distribution (routine runs) and Component 2 is the broader ranged distribution corresponding to training runs.
    2. **Mean Lower Bound:** $\mu_2 > 14$km, enforcing that Component 2 represents the longer runs.
    3. **Minimum non-zero σ_1:** $\sigma_1 > 0.1$ was enforced to aid in convergence by stopping σ_1 crashing to zero.

Note that while these priors make sense in terms of matching prior knowledge, and the constraints fit knowledge of the modelling we are trying to achieve, the values given above did come from a series of trial and error to find settings that allowed the MCMC sampling to converge. E.g. initially I applied a constraint on the means $\mu_1 < \mu_2$ for identifiability, but this failed to converge due to the variances swapping between iterations as the distributions had so much overlap.

2.3. Data Preprocessing

The raw activity data was filtered to include only 'Run' activities with a positive distance. The `Activity.Date` variable, originally stored as a character string, was converted to the `POSIXct` date-time format to facilitate time-series analysis in the subsequent results section. Additionally, the distance variable was stored as a character ‘chr’ and needed to be converted to a double for use in the model.

3. Results

3.1. Convergence and Parameter Estimation

An MCMC sampler implemented using the package Stan in R was used to draw samples from the posterior distribution.

The MCMC sampler was run for 4,000 iterations across four chains (2,000 warmup). All posterior distributions converged successfully, indicated by all $\hat{R}$ values equalling 1.00 and n_eff all greater than 5000.

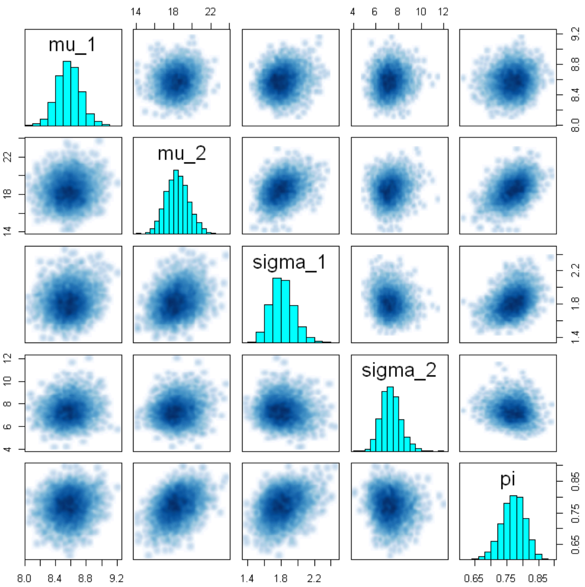

Figure 1: Pairs plot of the MCMC sampled parameters, showing unimodal gaussian distributions and an absence of collinearity. Good evidence of convergence.

The posterior means and 95% Credible Intervals (CI) from the original model are summarized below:

Parameter	Interpretation	Posterior Mean	95% CI
μ_1	Mean of Routine Runs (km)	**8.58**	[8.27, 8.89]
σ_1	SD of Routine Runs (km)	**1.82**	[1.57, 2.12]
μ_2	Mean of Training Runs (km)	**18.41**	[16.06, 20.89]
σ_2	SD of Training Runs (km)	**7.32**	[5.82, 9.18]
π	Proportion of Routine Runs	**0.77**	[0.69, 0.84]

The results show a clear distinction between the two behavioural groups: Component 1 is centered at **8.58 km** with low variability, while Component 2 is centered at **18.41 km** with a significantly higher standard deviation, confirming the initial hypothesis. The routine runs (μ_1) constitute approximately **77%** of all recorded activities.

Figure 2 below shows the mixture model and the two components that make it up compared with the raw data.

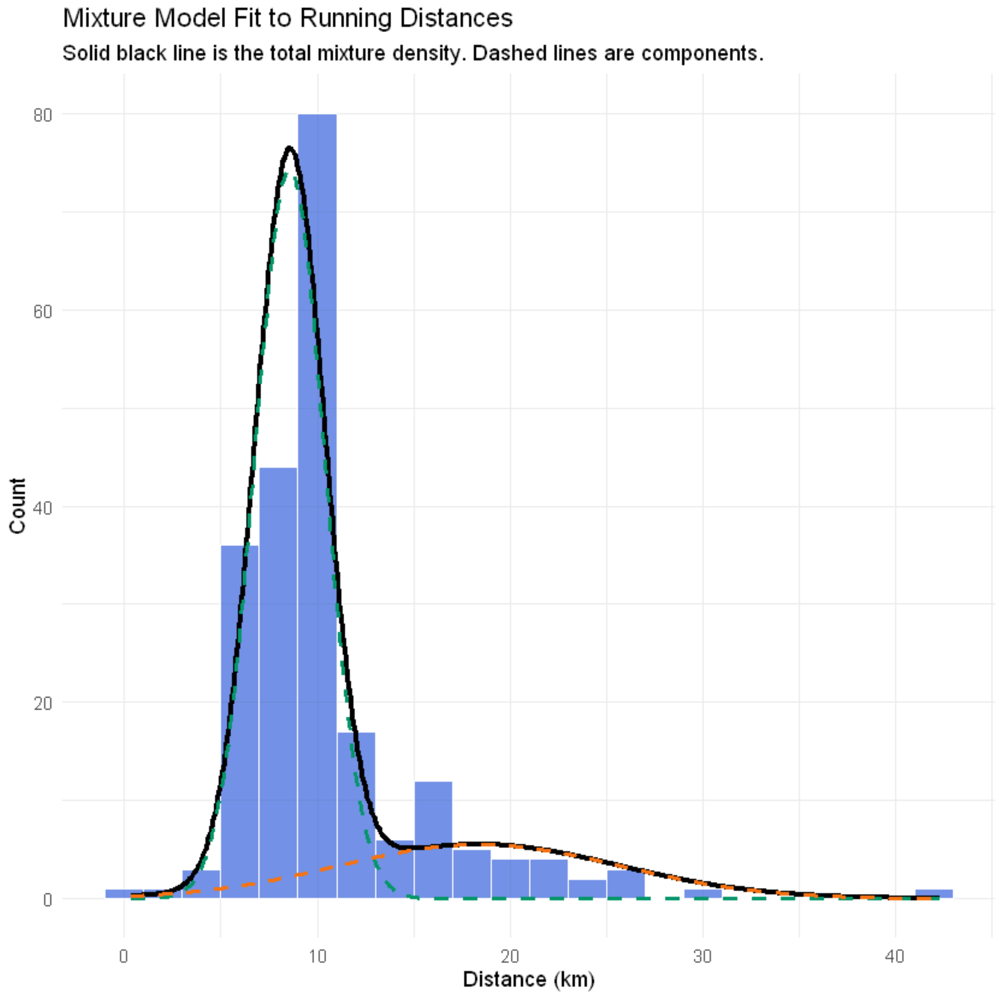

Figure 2: Mixture model and components scaled to match, and compared against, raw activity data, showing a good fit.

3.2. Prior Sensitivity Analysis

To test the robustness of the results, the model was refitted sequentially using two sets of progressively **weakly informative priors** by increasing the prior standard deviations for μ_1 and μ_2. This process determines the stability of the posterior distribution as the influence of the prior is reduced.

The comparison of key posterior estimates with 95% CIs is shown below:

Parameter	**Original Model** $SD_{\mu_1} = 1.5, SD_{\mu_2} = 2$	**Weak Prior** $SD_{\mu_1} = 3, SD_{\mu_2} = 4$	**Very Weak Prior** $SD_{\mu_1} = 3, SD_{\mu_2} = 8$
π (Routine Proportion)	0.77 [0.69, 0.84]	0.74 [0.65, 0.82]	0.73 [0.64, 0.81]
μ_1 (Routine Mean km)	8.57 [8.28, 8.88]	8.56 [8.25, 8.87]	8.55 [8.25, 8.86]
σ_1 (Routine SD km)	1.82 [1.57, 2.12]	1.74 [1.49, 2.02]	1.72 [1.47, 2.01]
μ_2 (Training Mean km)	18.41 [16.07, 20.89]	16.33 [14.36, 18.79]	15.71 [14.15, 18.07]
σ_2 (Training SD km)	7.32 [5.78, 9.17]	7.23 [5.95, 8.84]	7.22 [5.95, 8.74]

The parameters π, μ_1, σ_1 and σ_2 demonstrate excellent robustness across all three models, confirming that the data strongly defines the routine run habits and the variability of the training component.

The posterior mean for μ_2 exhibits a significant shift from 18.41 km (SD=2) to 16.33 km (SD=4), and then stabilizes at 15.71 km (SD=8). This dependence on the prior hints at this parameter getting less certainty from the data, which also shows up in the wide CI. This is because training runs are more dispersed and fewer in number, giving the data less power to overcome prior influence

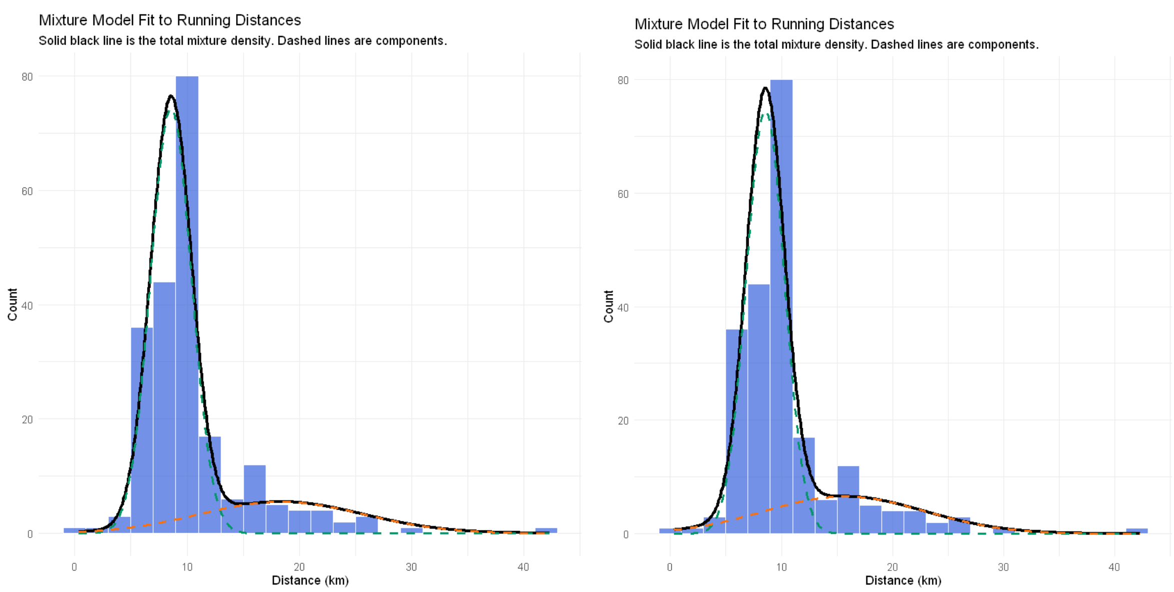

Figure 3: For comparison, the original model on the left vs. the "Very Weak Prior" model on the right. We can see the difference in the fit is minimal.

3.3. Posterior Predictive Checks

To assess model fit, posterior predictive checks were performed by generating 1,000 synthetic datasets from the posterior distribution and comparing them to the observed data. Figure 4 shows the observed data density (black) overlaid with 50 randomly sampled posterior predictive densities (blue). The observed data falls well within the range of predicted distributions, indicating good model fit.

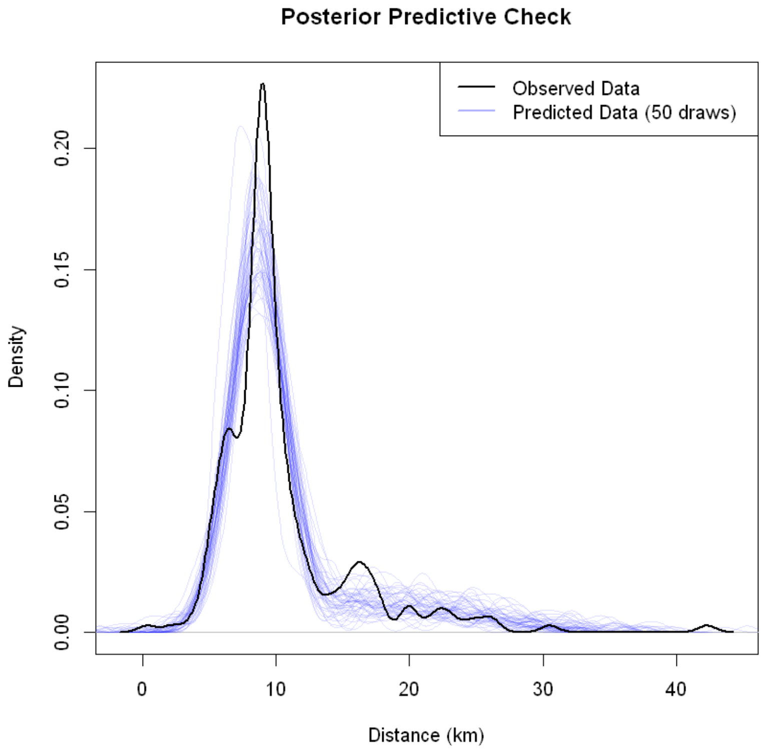

Figure 4: Observed data density (black) overlaid with 50 randomly sampled posterior predictive densities (blue).

3.4. Classification and Prediction

The model was used to classify each run in the dataset and also determine the distance threshold that separates the two groups. The threshold distance is the point where the probability of belonging to Component 1 equals the probability of belonging to Component 2 ($P(k = 1|Y) = P(k = 2|Y) = 0.5$).

To find this threshold we look for $P(k = 1|Y) - P(k = 2|Y) = 0$, which was achieved numerically using the R function uniroot, applied to the logs of the two distributions to avoid computational underflow. To find the 95% credible interval, the same threshold function was applied to all MCMC posterior draws and a CI interval found.

This found the classification threshold to be **12.95 km** with 95% CI [12.16, 13.87].

This distance serves as the decision boundary: runs below 12.95 km are most probable to be Routine, and runs above are most probable to be Training.

3.5. Training Timeline

Applying this classification to the raw data provides a visual timeline of training.

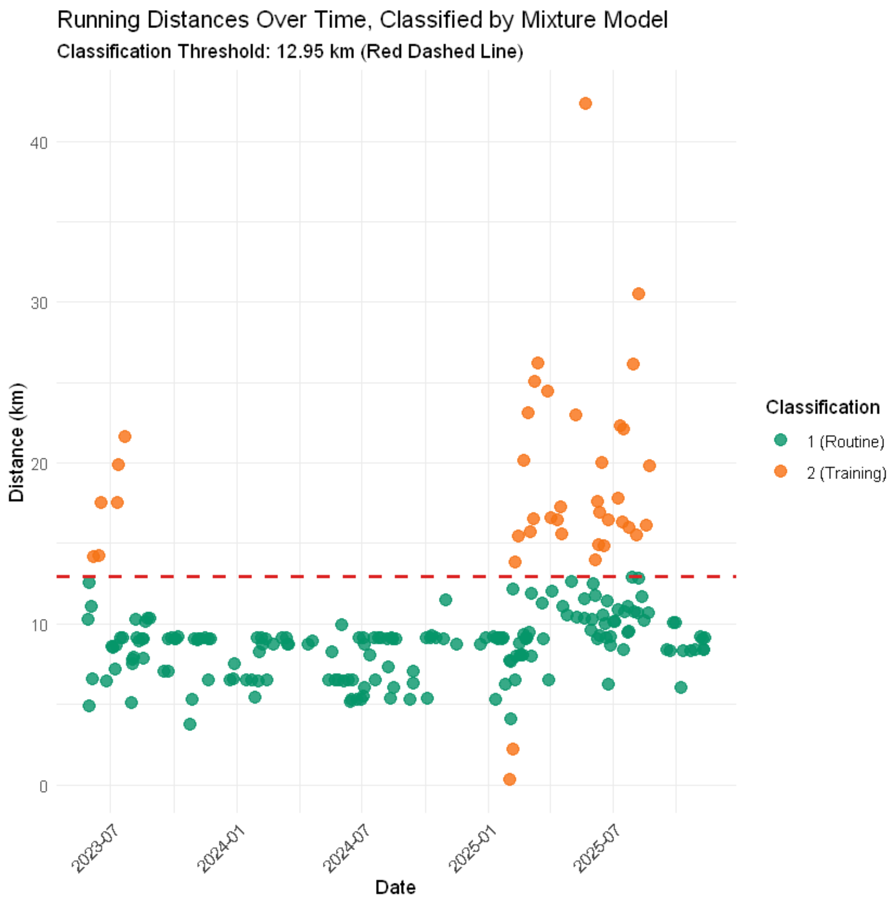

Figure 5: Classification of run activities in the raw dataset using the mixture model

The visualization shows clusters of "Training" (long) runs, likely corresponding to specific race preparation cycles, interspersed with long periods dominated by "Routine" runs. Note the classification of two shorter runs as “Training” as the broader “Training” distribution once again had a higher probability than the narrow “Routine” distribution - while not fitting with the desired outcomes here, a similar lower threshold could be found too.

4. Discussion and Conclusions

4.1. Implications of Findings

The mixture model shows that the author's prior knowledge that his running activity tends to be split into two regimes can be quantified into the two components of a Gaussian mixture model. This can then be used to classify run activities, and predict what regime the runner is in from the distances of runs they're doing.

The model was then used to find a predictive threshold value (12.95km) between the two regimes to aid in classification, and a 95% credible interval of that threshold was found to be quite narrow (less than 1 km above and below) giving good confidence in the value. This threshold was used to classify the existing runs, and serves as a predictive tool that could be used to classify future runs and give an indication whether a period of training had been entered into.

An interesting extension of this would be to re-fit the model with newer data and see how the parameter values change, and whether that gives indications about the runner's changing fitness. Another interesting extension would be to extend the model to multiple runners, applying a hierarchical mixture model.

4.2. Suitability of the Bayesian Mixture Model

The Bayesian mixture model was suitable for this task for several reasons:

- **Incorporation of Domain Knowledge:** Prior knowledge of two running regimes suggested a mixture model was appropriate, and the use of informative priors (μ_k centered on expected values) and the identifiability constraints ($\sigma_1 < \sigma_2$) allowed the integration of prior knowledge about running habits, leading to meaningful parameter estimates that resisted label switching.
- **Uncertainty Quantification:** The method automatically provides credible intervals for all parameters (e.g., π is estimated to be between 69% and 84%), giving a full quantification of the uncertainty in the training habit.
- **Framework for Future Data:** The Bayesian method is well suited for incorporating new data and updating the posterior. The posterior distributions obtained in this modeling work are ready to be used as a prior when new data is obtained.

4.3. Concerns and Reliability

Concerns Regarding the Dataset: There are minimal concerns about the dataset, as Strava has provided clean data, and the only concern would be that some points may not have been recorded accurately by the runner - in particular it is suspected the two particularly short runs were short due to running watch malfunction.

Model: The model assumes Normal components, and these are effective at providing a good fit for the model, however there is a concern that Normal distributions are non-zero for distances < 0 which is physically impossible. Given the μ_k and σ_k of the posterior distributions this is of minimal concern as the probabilities are very low, however other distributions such as log-normal, gamma or truncated normal would avoid the issue. Because of this I tried using a log-normal mixture model for comparison, but was unable to get it to converge well - see section 6 of the Jupyter notebook for further details. This may have been because the data could be well fitted by a single (i.e. non-mixture) log-normal model - however this wouldn't be useful for the routine vs. training objective.

Reliability: The strong separation of the posterior distributions for μ_1 and μ_2 and the good convergence diagnostics indicate that the estimated parameters are highly reliable within the context of the chosen Normal mixture structure. The posterior predictive checks show the observed data sits well amongst the predicted datasets, showing good reliability. The computed threshold also shows reliability given the relatively narrow 95% CI.

5. Bibliography

- Strava Activities Dataset for Warwick Simpson, obtained from strava.com on 26/11/2025
- Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., & Rubin, D. B. (2013). *Bayesian Data Analysis*. Chapman and Hall/CRC. (Reference for Bayesian Methodology)